

FINAL EXAMINATION

(Notes: Laptops and mobile devices are not allowed)

Question 1:

Draw the ER diagram for the following situations, and indicate keys for entity sets.

Entity sets Leagues, Teams, and Players. League names are unique. No league has two teams with the same name. No teams has two players with the same number. However, there can be players with the same number on different teams, and there can be teams with the same name in different leagues.

Question 2:

Draw the Relational database schema for the ABC Airline which is described as follows, Each FLIGHT is identified by a flight number, and consists of one or more FLIGHT_LEG with leg number 1, 2, 3, and so on. Each FLIGHT_LEG has scheduled arrival and departure times, airports, and one or more LEG_INSTANCES, one for each date on which the flight travels. For each LEG_INSTANCES, the airplane used on the leg, the actual arrival and departure times, the airports and the number of seat reservations are kept. An AIRPLANE is identified by an airplane id and is of a particular airplane type. An AIRPORT is identified by an airport code.

Question 3:

Given a relation whose schema is Product(maker, model, type) and its instance as the right hand side. Write an expression of relational algebra to find the makers who sell exactly two different models of “laptop”.

maker	model	type
A	100	laptop
A	200	tivi
A	300	printer
B	104	laptop
B	207	tivi
C	107	laptop
C	109	laptop

Question 4:

Given the database schema,

Student(Name, BirthDate, BatchOf, Faculty)

Course(Name, Semester, Year, Instructor, Faculty)

Register(SName, CName, Grade)

Prerequisite(CName, PreCName)

Write an expression of tuple relational calculus to find students who register courses of their faculty.

Question 5:

Given two schemas :

Car(CarID, CarName, Type, ManufacturerID, Year)

Manufacturer(ManufacturerID, ManufacturerName)

Describe the following integrity constraint: After 1970, Toyota started to manufacture the type of 4-seat cars.

The context :

The content :

Table of inference :

Question 6:

Given the schema Q, where keys are A and H,

Q (ABCDEFGHIJKL)

And a set of functional dependencies F, as follows,

f1: A $\rightarrow$ BCDEFHL

f2: AE $\rightarrow$ H

f3: HB $\rightarrow$ C

f4: AB $\rightarrow$ HJ

f5: B $\rightarrow$ LK

f6: E $\rightarrow$ FG

f7: H $\rightarrow$ IJBA

(a) Does the functional dependency AE $\rightarrow$ K belong to F? Brief explanation

(b) Evaluate the normal form of this schema.

Question 7:

Given the schedule S as follow,

$$S = w1(X) w2(X) w3(X) w4(X) w5(X) r4(Y) w1(Y)$$

(a) Draw the Precedence Graph of S, P(S).

(b) Is the schedule S conflict-serializable? Brief explanation.

(c) If S is conflict-serializable, how many serial executions may the schedule S have? List them below. If S is not conflict-serializable, how many cycles may P(S) have? List them below.

Question 8:

Given the schedule S as follows,

S	T1	T2	T3	T4
1	Rlock A			
2		Rlock A		
3	Unlock A			
4			Wlock B	
5		Unrock A		
6				Wlock A
7			Unlock B	
8	Rlock B			
9				Wlock C
10	Unlock B			
11		Wlock B		
12				Unlock A
13			Wlock A	
14		Unlock B		
15			Unlock A	
16				Unlock C

Is the schedule S converted to a conflict-equivalent serial schedule? Brief explanation.

Question 9:

Consider the following transaction,

- 1 <start ckpt (T1, T2, T3)>
 - 2 <T1, A, 10>
 - 3 <commit T1>
 - 4 <T2, B, 20>
 - 5 <abort T2>
 - 6 <T3, C, 30>
 - 7 <end ckpt>
 - 8 <commit T3>
- end of transaction log –

(a) The following transaction log is created by Undo or Redo method? Brief explanation.

(b) Let recover the above database. Brief explanation.

--The end--